

Fig. 3: The components library in QUCS

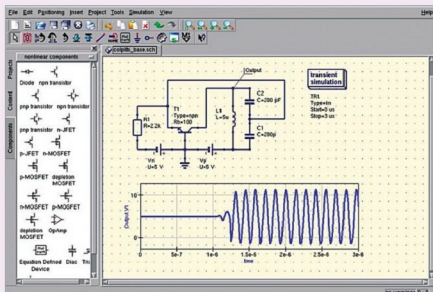


Fig. 4: The QUCS UI with a simulation graph

bridges among others. The library is quite large and can be further expanded by the user. The text editor displays and records different logging information and edit component files. There are in-built tools in the software that can generate different electronic components. For instance, a filter synthesis application can be used to generate various circuit filters, transmission line calculator can design and analyse newly created transmission lines, attenuator synthesis application can create passive attenuators, and so on.

QUCS can import and export elements like netlists, datasets and data tools from other EDA or CAD software. This is done by the command line conversion program tool. It can generate digital simulations using VHDL from external tools like FreeHDL. It can also connect to a software called ASCO for circuit optimisation.

Getting started

Installation of QUCS is simple and well-instructed. Once it is installed, launch it to start a new project. Provide the project name and confirm to

be directed to the workspace. There, you can add your components and wires through pick-n-place applications. Select the desired components from Components folder in the panel. Right-clicking the circuit elements enables you to perform further operations like editing the component specifications or inserting relevant text labels.

Once you have created the circuit, go to Simulation in the menus and click Simulate. A new window will open, which allows you to choose the type of visual graph for simulation data. Select the graph of your choice and simulate the circuit to see the different data points and circuit behaviour.

The QUCS library

The latest release of QUCS library is quite versatile and contains all components needed to build different kinds of circuits. It contains sources, probes, transmission lines, digital components, lumped components like R, L, C amplifiers, phase shifters, non-linear components like diodes and transistors, file containers like SPICE netlists, paintings and so on. Users can include various transistor models like JFET, MOSFET, SGP, various HICUM models, MESFET among others.

Dataset items include AC/DC voltage and current, noise voltage, noise current, transient voltage, transient current and S-parameter value. Types of charts include Smith, Cartesian, Tabular, Polar, Smith-Polar combination, 3D-Cartesian, Locus Curve, Timing Diagram and Truth Table.

QUCS offers a range of expansion and customisation options. External tool plugin option is quite useful. Operations like harmonics analysis, circuit optimisation, transient analysis and sub-circuit simulation can be readily performed using QUCS. The software has a powerful data post-processing capacity and support for multiple languages. All this makes QUCS a very useful circuit simulation tool. **EFY**